Enhance Your Programming Experience with IPython

Take learning, teaching and computing with Python to a new level with this interactive computing environment.

Software development is a nuanced activity. Unless you are doing legacy product maintenance—where there are severe restrictions on creativity—you are really learning language mechanisms, trying out algorithms, tweaking computation models, or sharing a couple of programming recipes with your juniors. This is no surprise to those who have worked with formal iterative development cycles or to students and scientists wrestling with data.

Python is a popular language to learn and teach, and of course to program in. So, is there an environment that exposes the potential of Python in a fun manner and also creates an ecosystem for both teachers and learners?

IPython is your answer. It can be your favourite learning, teaching or computing environment depending on what your primary purpose is. What you learn and develop in IPython can really contribute to highly productive sessions with a formal IDE in the final stage of a development phase.

Assuming you have Python already installed, install *ipython* and *ipython-notebook* using your package manager, and you are ready to go. Run the following command:

```
$ ipython notebook
```

Your browser should open with a notebook dashboard where you can start a new notebook or open an existing one. (If you are running Python 3, the recommended download is *ipython3*.)

Intuitive interaction

A typical open notebook with code and output could look like what's shown in Figure 1. IPython is very well documented and the place to learn more about this exciting tool is the *ipython.org* website. But before you get there, let us look at some basic usage that will take you well past the 'Hello world' stage.

IPython runs code in cells. A cell is a container for